

Name of the Student
Reg. No
GitHub Link

SIMATS ENGINEERING
CO2 - ASSESSMENT TOOL 3
Class Test

18/8/20

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE: MLA04
Slot : D

COURSE OUTCOME COVERED

CO 2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage : 15%

Assessment Tool Weightage: 20%

Duration: 90 minutes

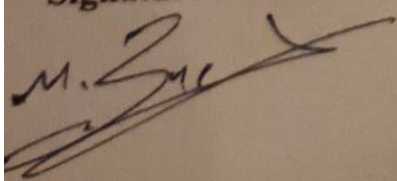
Marks: 25

Code of Conduct:

I Coppo Syc Kumar (192325121) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student:

Coppo Syc Kumar

11/2/20

Q. Machine Learning vs Deep Learning

Machine Learning:- It is a branch of AI where algorithm learn from patterns from data.

Deep Learning:- It is a subset of machine learning that uses multi-layer neural networks.

Aspect	Machine Learning	Deep Learning
Architecture	Decision Tree, SVM, KNN	Neural Network with many hidden layers.
Data Requirement	works with small/medium datasets	usually requires large datasets.
Feature extraction	Requires less computational power	Requires more computational power after GPUs
Training	usually faster	usually slower
Application	structured/tabular data	Images, speech, videos, NLP

Machine Learning Applications:- Loan Approval, Spam Detection

Deep Learning Applications:- Face recognition, self-driving vehicles

Q. Representation learning, width and Depth.

Representation learning:- It is a technique where a neural network automatically learns useful features from raw data.

Ex: Image $\rightarrow$ Edges $\rightarrow$ shapes $\rightarrow$ objects $\rightarrow$ classification.

Width:-

width means the number of neurons in a layer

Input $\rightarrow \{ \dots \} \rightarrow$ output

more width

- * can learn more features.
- * increases computations cost.
- * can improve performance.
- * Too much width may cause overfitting.

Depth:-

Depth means the number of layers in a neural network

Input $\rightarrow$ layer-1 $\rightarrow$ layer-2 $\rightarrow$ layer-3 $\rightarrow$ output.

more depth

learns complex hierarchical features.

- $\rightarrow$ Increases computational complexity.
- $\rightarrow$ can improve performance.

Example:-

Pixels $\rightarrow$ Edges $\rightarrow$ shapes $\rightarrow$ objects.

Q3 Relu, Leaky Relu and ELU

Activation functions introduce non-linearity into neural network.

Relu :- $f(x) = \max(0, x)$

- Positive input $\rightarrow$ output is x
- Negative input $\rightarrow$ output is 0

Advantages :-

Simple & fast
Easy to compute

Disadvantages

• can suffer from the dying ReLU problems

Leaky Relu :- $f(x) = \begin{cases} x, & x > 0 \\ \lambda x, & x \leq 0 \end{cases}$

where $\lambda \rightarrow$ is a small value such as 0.01
unlike ReLU, it allows problems

advantages

• Reduce dying ReLU problems

Disadvantages

Requires choosing λ

ELU :- $f(x) = \begin{cases} x, & x > 0 \\ \lambda(e^x - 1), & x \leq 0 \end{cases}$

Advantages :-

smooth negative region
Reduce dying neuron problem
can improve optimization

Disadvantages

• more computationally expensive than ReLU

Qn Unsupervised learning RBM and Autoencoder

unsupervised learning: It learns pattern from unlabeled data.
ex Images $\rightarrow$ Neural Network $\rightarrow$ learned features / patterns.

Applications include:

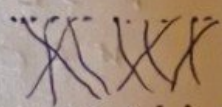
- Feature extraction
- Data compression
- Anomaly detection
- Clustering.

Restricted Boltzmann Machine (RBM)

RBM is a neural containing:

- visible layer
- Hidden layer.

visible layer



Hidden layer

There are no connection between neurons in the same layer.

working:

- * $\rightarrow$ Input is given to the visible layer.
- * $\rightarrow$ Hidden neurons learn patterns from the input.
- * $\rightarrow$ The model learns the probability distribution of data.
- * $\rightarrow$ It can reconstruct or generate similar patterns.

Applications:

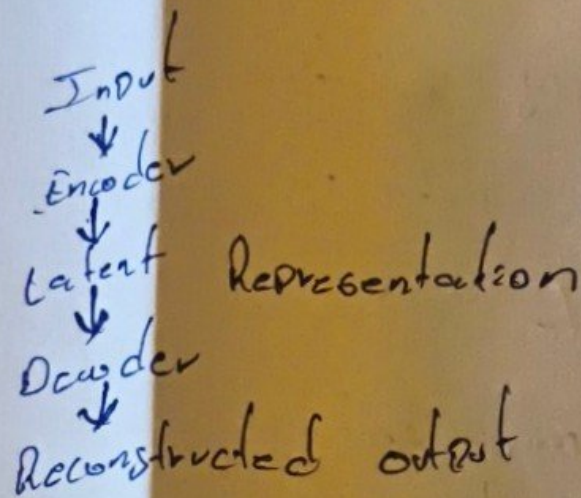
- Recommendation system.
- Feature learning.
- Generative modeling.

MLA0405 - Deep Learning

Q5 Autoencoder and its uses :-

An autoencoder is an unsupervised neural network that learns to compress input data and reconstruct

Architecture &



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Encoder

converts input into a smaller representation.

$$z = f(x)$$

2. Latent Space

contains the important features of the input.

3. Decoder

Reconstructs the original input:

$$\hat{x} = g(z)$$

The model tries to make:

$$x \approx \hat{x}$$

uses of Autoencoders.

- * Dimensionality Reduction
- * Data compression.
- * Image Denoising.
- * Anomaly Detection
- * Feature extraction

Example

1000 feature



Encoder



50 features



Decoder



1000 Reconstructed features.